

ELE432 Preliminary Work 3 Multi-Cycle RISC-V Processor

1) I spent total of 5 hours on this preliminary work. **Note:** All of the codes are in GitHub.

2)

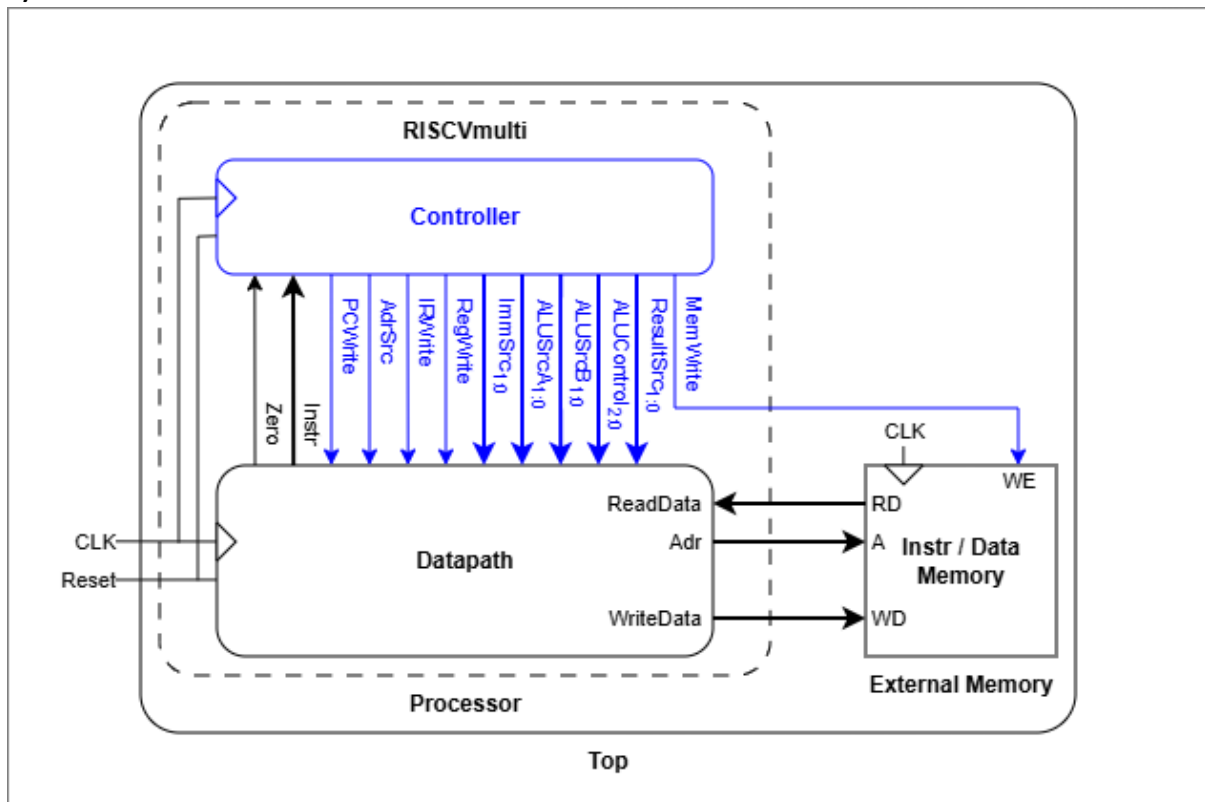


Figure 1

3) RTL Schematic showing the Top-Level Processor Module and the Processor:

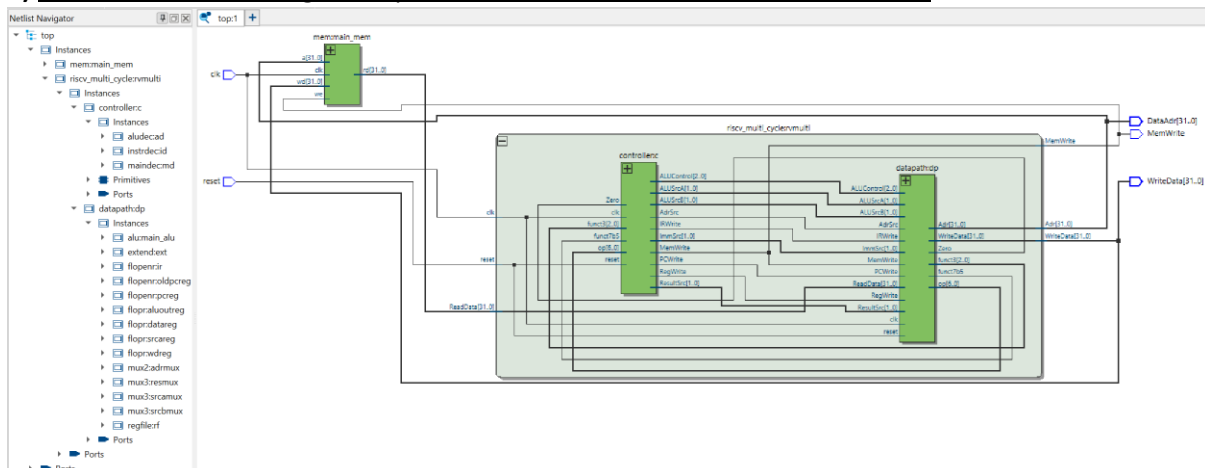


Figure 2

4)

Table 1: Expected Operation (after two cycles of reset)

Step	PC	Instr	State	Result (Hex)	Result Notes
3	00	00500113	S0: Fetch	04	PC+4
4	04	“ “	S1: Decode	X	OldPC+Immediate
5	04	“ “	S8: Executcel	X	ALUResult = x0 (0) + 5 = 5
6	04	“ “	S7: ALUWB	05	Result = ALUResult
7	04	“ “	S0: Fetch	08	PC+4
8	08	00C00193	S1: Decode	X	OldPC+Immediate
9	08	“ “	S8: Executcel	X	ALUResult = x0 (0) + 12 = 12
10	08	“ “	S7: ALUWB	0C	Result = ALUResult
11	08	“ “	S0: Fetch	0C	PC+4
12	0C	FF718393	S1: Decode	X	OldPC+Immediate
13	0C	“ “	S8: Executcel	X	ALUResult = x3 (12) + (-9) = 3
14	0C	“ “	S7: ALUWB	03	Result = ALUResult
15	0C	“ “	S0: Fetch	10	PC+4

...and so on until the progam goes into a loop at PC = 0x50.

5) My system passes the testbench.

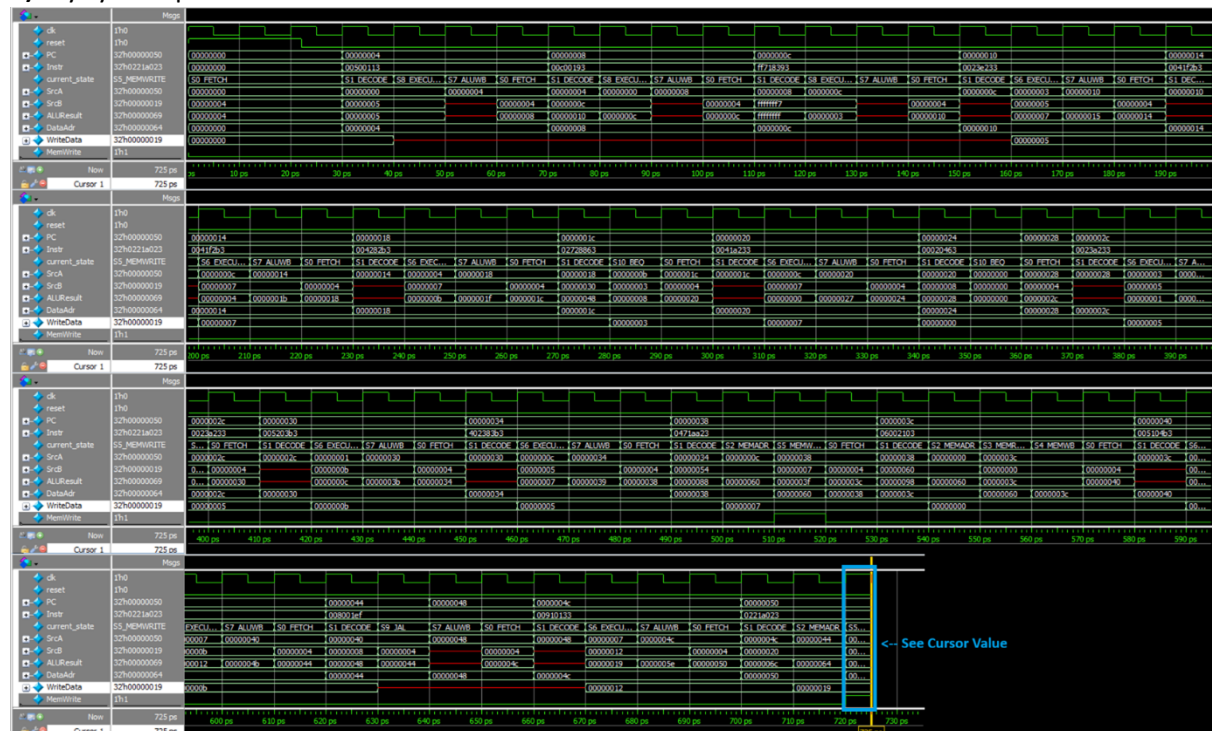


Figure 3

Questa Transcript Output:

```
run -all
# Simulation succeeded
# ** Note: $stop : C:/Projects/quartus/PRE3_Multi-Cycle_RISC-V_Processor/riscv_testbench.sv(30)
# Time: 725 ps Iteration: 1 Instance: /testbench
# Break in Module testbench at C:/Projects/quartus/PRE3_Multi-Cycle_RISC-V_Processor/riscv_testbench.sv line 30
```

6) GitHub Link: https://github.com/20egemen04/Multi-Cycle_RISC-V_Processor